

Technical Stack

Languages & Frameworks: Python (Jupyter, Dash, Pandas, Numpy, SciPy), Java (Micronaut, Spring), Clojure, Bash

Data & Infrastructure: PostgreSQL, BigQuery, Google Logging Analytics, Splunk, Spark SQL, Google Cloud Platform

Visualization Frameworks: Looker, Splunk, Dash, Shiny

Other: Vertex AI (Gemini), OpenAI, Scikit-learn, TensorFlow

Professional Experience

Broadcom Inc. (Rally Software)

Senior, Staff, Principal Data Engineer & Data Scientist

Broomfield, CO

Oct 2015 – Oct 2025

- Led technical and algorithmic sides of development for Agile flow metrics and Agile forecasting products.
- Developed a "Smart" Cumulative Flow Diagram for analyzing team delivery metrics.
- Led team and train-level portfolio flow metrics development.
- Used Vertex AI (Gemini) to produce flow metrics chart interpretations and coaching advice for teams.
- Led capacity and delivery forecasting development for team and train-level planning.
- Helped develop and maintain a distributed tracing infrastructure for a visible and reliable usage and performance metrics system to report 100% of user interactions in a SaaS environment involving 250K+ users. Used the resulting data with Splunk, BigQuery, Looker, Spark, and other systems to analyze interactions.
- Built and maintained a "product data hub" for those within Rally to track feature adoption, subscription usage, performance, and user input.
- Built a leading indicator for subscription renewal predictions using usage data and random forests.
- Worked in Python, Java, Clojure, Bash, Scala, JavaScript (React), and elsewhere as needed.

SpotXChange

Data Scientist

Broomfield, CO

Apr 2014 – Sep 2015

- Developed a large-scale Hive datastore for 600+ frontend ad-tech bidding servers.
- Tracked automated view bidding for website advertisements.
- Led development of fraud activity investigations using big data, Hadoop, ETL, MySQL, and Tableau.
- Built Tableau dashboards for product and C-level insights.

Silicon Graphics, Inc. (SGI)

Software Engineer

Longmont, CO

Jun 2013 – Mar 2014

- Led development and testing of SCSI Enclosure Service (SES) Python/Cython module.
- Worked on various firmware/control modules in C, C++ (Boost), Python, and Cython.

Education

Ph.D. in Mathematics (ABD/2014)

University of Colorado–Boulder (Alexander Hulpke and Keith Kearnes)

- Computational group and graph theory, permutation group algorithms, NCAR & XSEDE supercomputers
- Developed a massively parallel approach to solving partition backtrack problems in group and graph theory.

M.A. in Mathematics – 2010

University of Colorado–Boulder (Keith Kearnes)

- Computational group theory, representation theory

M.S. in Mathematics – 2006

University of Vermont (David S. Dummit)

- Computational algebraic number theory
- Developed an algorithm for finding number fields satisfying given properties and having minimum discriminant. Found the first known instance of a totally real quintic number field having minimal signature group rank.

B.S. in Mathematics – 2004

University of Michigan–Flint

Miscellaneous

Volunteering

- In my spare time (most recently in December of 2025), I volunteer with the Save Elephant Foundation in Northern Thailand to help rescue and care for abused elephants. I'm also Santa for roughly 250 Thai children.

Meetups and Knowledge Sharing

- I've spoken four times at local Agile Meetup groups, discussing Cumulative Flow Diagrams (CFDs), probability and group decisions, Monte Carlo simulations for Agile forecasting, and some interesting findings from Little's Law.
- Since resigning from Rally, my goal is to share some Agile metrics knowledge, specifically related to Little's Law and forecasting (new things beyond the scope of my work at Rally), via my website: www.jasonbhill.com.

Other Employment

- In addition to teaching many courses in mathematics while attending graduate school, I've held two adjunct instructor positions, one at Saint Michael's College in mathematics and one at The Community College of Vermont in photography.
- I've worked multiple times as a contract photographer for Automobile Magazine.